

JOSHUA PLUMMER • LECTURER IN MATHEMATICS

Accessible and Rigorous

Designing Mathematics Instruction for the Modern Learner

For instructors teaching gateway mathematics in higher education
College Algebra • Pre-Calculus • Elementary Statistics

What’s Inside

Seven parts, built in sequence. Parts 1 through 3 establish the people, relationship, and problem. Part 4 contains three independent working sessions. Parts 5 through 7 address tools, continuation, and the stance holding the design together.

The three sessions in Part 4 stand alone. Each begins with a current course problem and ends with a usable artifact. Enter with the session that matches the pressure you already feel.

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Teal sidebars Fast orientation, decision points, and implementation checks.	Gold callouts Claims or cautions worth carrying into practice.	Working pages Printable planning, review, and reflection tools.
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Worksheet Index

The working pages are collected at the end so they can be printed, duplicated, or opened during a coaching session without interrupting the main argument.

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PART ONE

Who This Is For

The Audience

Gateway mathematics instructors at regional, teaching-focused universities.

Adjuncts, full-time lecturers, and tenure-track faculty teaching College Algebra, Pre-Calculus, and Elementary Statistics, often with co-requisite support that pairs college-level instruction with embedded academic help.

Nearly all were trained as experts in their discipline rather than in how adults learn. Most hold graduate degrees in mathematics or a closely related field, and formal coursework in instructional design is rare. A substantial share teach part time, often across multiple institutions, and full-time lecturers carry heavy loads. Time for professional learning is scarce, and anything that takes it has to return something usable.

The students are largely non-traditional working adults with uneven preparation, which leaves instructors little patience for training that ignores what the classroom is actually like.

Where this audience differs most

Not in mathematics. In how far each person has moved toward accessible, student-centered teaching (El-Amin, 2020). Newer instructors often carry fewer entrenched habits and adopt these methods readily. Veterans more often prefer traditional approaches they genuinely believe serve students better, and that belief deserves to be taken seriously rather than argued away.

What the setting allows, and what it does not

This is a high-autonomy environment in reputation and a more constrained one in practice. Faculty exercise broad independence over how they teach, and professional development is almost entirely voluntary, so no compliance structure can require anyone to adopt a new design practice.

At the same time, gateway courses are usually coordinated through common syllabi, shared pacing, and departmental exams. An individual instructor's control over content and assessment is partly surrendered to the department. Everything in this toolkit is therefore scoped to design moves an instructor can actually make inside a coordinated course.

AT A GLANCE

Who

Adjuncts, lecturers, and professors teaching gateway mathematics.

Two constraints

Professional learning is voluntary; coordinated courses limit what one instructor can change alone.

Best entry point

A problem the instructor already recognizes in a live course.

Readiness signals

Skipped course pages, strong homework with weak exams, repeated reteaching, or outcomes under review.

Design implication

Offer relevant choices with enough scaffolding to finish something useful.

Choice supports agency only when the options still serve the learning goal.

Two Problems

Both are design problems, not content problems.

Gateway courses often assume conditions our students do not have. They assume attendance, preparation, and available time. Many of our students have none of the three. Gateway courses nationally produce disproportionately high rates of failing and withdrawal grades and act as a primary barrier to degree completion, particularly for historically underserved students (Koch, 2017).

A correct answer no longer proves mathematical reasoning. A model produces that answer in seconds.

Preserving the integrity of an assessment now means redesigning it rather than policing misuse after the fact (Cotton et al., 2024; Tertiary Education Quality and Standards Agency, 2025). Prohibiting AI in a syllabus does not change what an assignment is capable of detecting.

Neither is solved by knowing more mathematics. Both are visible, both are consequential, and both belong to the design rather than to the students.

What You Already Bring

This starts from what you know.

Content knowledge, built over years of explaining the same ideas to people who did not understand them the first time.

Diagnostic instinct. You know which form of an equation produces the most errors, which habits students carry in from high school, and which topics they dread. That knowledge is granular and it is not written down anywhere.

Platform fluency. You already navigate Canvas, post materials, and manage a course shell.

Those three add up to a fourth: **you can redesign a course quickly.** That capacity is real, and it is conditional on one thing – trusting that the redesign addresses a gap you actually recognize in your own teaching. What is missing is not mathematics or Canvas. It is a structure to put them together in.

What Adults Need From Professional Learning

Six conditions, and what each one means for this audience.

A reason

Adults will not invest real effort until they understand why the learning matters (Knowles et al., 2015). Because engaging this toolkit is voluntary and nobody is compelled to change, the professional rationale has to come before any content begins.

Ownership

Treating adults like trainees produces artificial compliance and no real engagement (Knowles et al., 2015). There is an added risk here: an instructor whose identity rests on content expertise may experience the role of novice in instructional design as a threat rather than a starting point. Every module is therefore framed as a professional choice, and the design work is scaffolded rather than assumed.

Their experience taken seriously

Experience is the adult learner's most valuable resource, and habits formed from that experience are also a source of resistance (Knowles et al., 2015).

You have refined lesson plans, watched students struggle, and tested many classroom methods. That same history can lock in a lecture-heavy approach and the belief that traditional methods are the only route to rigor (El-Amin, 2020). So the work starts with before-and-after redesigns of your own materials, which puts your experience to work rather than putting it on trial.

Timing

Adults are most ready to learn when their situation demands it (Knowles et al., 2015), and readiness arrives unevenly across this audience. An instructor whose DFW rates are under departmental review is far more ready than one whose courses seem fine and whose evaluations are glowing. There is also the instructor who has begun to suspect that aced homework and failed exams are connected. Enter wherever the pressure is.

A problem, not a subject

Adults learn to solve problems rather than to master subjects (Hagen & Park, 2016; Knowles et al., 2015). This audience will sit through a talk on theory, but what they adopt is a concrete design change they can try in a live course. Each session is organized around a single design move applied to a real assignment, so working through one produces something immediately usable.

Internal motivation

External motivators can start learning; only internal ones sustain it (Knowles et al., 2015). Instructors care about their students and about the integrity of their courses, and they also work under pressure from metrics like DFW rates (Koch, 2017). Nothing here appeals to administrative expectations, because doing so would convert the motivation the whole design depends on into something external.

One honest limit

The picture of the independent, self-directing adult learner is a partial one (Brookfield, 1995; Merriam & Bierema, 2014; St. Clair, 2024), and it understates something basic about gateway courses: they are coordinated, with common syllabi and department-issued exams that limit what any one instructor can change alone. And self-direction cannot be assumed across domains (St. Clair, 2024). An instructor may be entirely willing and still lack the pedagogical or technical knowledge to direct their own learning here. That is why every application above is paired with scaffolding rather than encouragement.

What Sustains Engagement

Autonomy, competence, relatedness – and what threatens each.

Adults who have autonomy, competence, and relatedness sustain engagement readily. The same motivation is undermined when external pressure is applied (Ryan & Deci, 2000). Both halves describe this audience.

Autonomy, and why it comes first

Instructors cannot be required to use any of this, and any implication of a mandate ends engagement before it begins. They are accustomed to full authority over their own courses and regard that authority as intrinsic to the job. So each part stands alone as far as possible, letting someone enter at whatever point matches their current concern rather than following a set sequence.

Every design move is presented as a professional choice with a rationale attached, never as a requirement.

Autonomy is the central need here. If it is violated, people disengage before competence and relatedness ever get a chance to operate.

Competence

Experts in one field dislike being made beginners in another and will disengage quickly when that is how it feels. Many also carry the memory of past training that asked a great deal and returned very little. The response is to build in early wins: scaffolded before-and-after examples of real assignments, and low-stakes design tasks finishable in one sitting. You leave each session having built an asset rather than having read a theory.

Relatedness

Teaching can be solitary. Adjuncts in particular may go years without a colleague ever seeing their materials. So the work is built around recognizing your own conditions in someone else's, through composite vignettes drawn from documented courses and published accounts of instructors making these changes voluntarily.

Three barriers worth naming

Time. Heavy loads, possibly across several institutions, departmental duties, and student-facing commitments.

Prior bad experience. Professional development that asked much and returned little affects both initial motivation and open-mindedness.

Pass-rate pressure. The most corrosive of the three, because it converts an intrinsic motive into a compliance-based external one (Ryan & Deci, 2000).

All three get the same answer: short, self-contained work that produces something immediately applicable, framed only in terms of the instructor's own goals.

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PART TWO

How This Coaching Works

Partnership, Not Evaluation

The distinction that governs everything else.

Adults are more likely to change when they feel a sense of agency over the process, and attempts to force change produce resistance (Knowles, 1984). Veteran instructors arrive as subject-matter experts who have invested years developing lesson plans and establishing how their classrooms run. Change therefore begins with reflecting on existing practice and comparing current outcomes to what the instructor hopes to achieve.

Faculty are already subject to student evaluations, peer observations, and annual reviews, so anything resembling judgment of their teaching invites defensiveness rather than openness. An evaluative relationship asks whether an instructor meets a standard. A coaching relationship asks what they want to accomplish and how you can help them get there (Knight, 2007).

You enter as a fellow practitioner sharing a design problem with a colleague, never as an expert auditing competence.

Honoring autonomy in practice

Your role is to bring a colleague's attention to relevant evidence while asking questions and offering options, and in doing so to let the instructor name the problem and set the goal. When you bring a design move, such as a reasoning-based problem set built to resist AI shortcuts, offer it as one option a colleague can adapt or decline (Knight, 2007). A mandated goal is not one the instructor owns.

Self-direction here is not a trait some instructors happen to have. It is a defining feature of how adults learn (Merriam, 2001), so room has to be left for it by design.

For an audience with thin professional development support and a reasonable skepticism toward being trained, a coaching stance built on allowing autonomy is both more respectful and more likely to produce a change in the work. If a colleague leaves with their own version instead of yours, the work succeeded.

AT A GLANCE

Stance

Partnership, not evaluation.

Coach brings

Questions, relevant evidence, options, and a willingness to learn.

Instructor keeps

The problem, the goal, the decision, and the final version.

Conversation is working when

The instructor does most of the sense-making, names the goal, and freely edits or declines the proposed move.

Avoid

Hidden reporting, compliance language, premature solutions, and goals chosen by the coach.

The next step should be specific enough to try and voluntary enough to own.

Five Commitments

What each one looks like, and what its absence looks like.

Partnership

Enter as a co-designer who shares the problem rather than an expert ready to resolve it. This puts you and the instructor on the same side, and when the relationship is built that way there is nothing to defend against (Knight, 2007).

In practice: instead of telling an instructor their videos need work, arrive with a problem of your own. “I can’t figure out why my videos on asymptotes aren’t being watched. Can we compare what we’re each doing?” The inquiry becomes mutual, which is what prompts a genuine answer.

Without it: the conversation turns into an audit, and the instructor spends it defending past choices instead of examining them.

Equality

The experiences, thoughts, and beliefs of the instructor are as valuable as the coach’s (Knight, 2007), and a competent coach listens to understand before responding. A coach who treats their own framework as the only valid lens signals that years of practice do not matter, and an instructor can withdraw from the relationship on the spot.

In practice: if a veteran tells you students skip the co-requisite session because it meets at 8 a.m., treat that as data that outranks any theory you came with, and build the work around it.

Without it: the learning runs one direction and the instructor’s knowledge of their own students is never invited into the conversation.

Empowerment

Coaching should empower the instructor to name their own problem and decide their own goal. Create space for their voice rather than filling it with your own; the aim is to help people find their voice, not to make them think a particular way (Knight, 2007).

In practice: avoid handing over a “use more active learning” directive. Ask what frustrates them most about how students engage with their materials, and begin where the answer takes you.

Without it: the coach’s priorities quietly replace the instructor’s.

Reflection

The instructor has to be free to make sense of what you are proposing in their own way, and to weigh an idea before accepting or rejecting it (Knight, 2007). An instructor handed a finished practice and told to implement it has been treated as a technician rather than a thinker with agency in their craft.

In practice: when you bring a reasoning-based problem set, do not ask them to use it. Walk them through the thinking behind it and leave room for them to decide what fits their course and their students. They might keep the reasoning prompts but rework them for another course, or set your version aside and build their own that accomplishes the same thing. The point is that they leave with their version, not yours.

Without it: the practice gets implemented exactly as handed over, and it is dropped the first time it does not fit their course.

Success

Success means the new outcomes meet the goals the instructor set (Knight, 2007, 2018). Reciprocity matters here too: expect to learn as much as you give (Knight, 2007).

In practice: the win is your colleague seeing more students complete a reasoning-based problem set on their own terms, and it is also you walking away with a classroom move you can use. If an instructor sees you learning from them, they know they are not being assessed.

Without it: success gets declared the moment a method is adopted, and nobody ever checks whether anything changed.

The First Conversation

Open, gather evidence, then let them set the goal.

Opening as a partner

State plainly that this is not an evaluation and that they decide where the agenda goes. Then ask one open question: if you could change one thing about how your students work through the course materials, what would it be? Letting them choose the direction is what puts you inside the problem with them rather than above it (Knight, 2018).

Gathering evidence

Impressions and assumptions get replaced by evidence from three sources.

Video. The instructor records one class segment or one screencast of a worked-example video, and you watch about five minutes together. Video matters because it lets the instructor see their own practice rather than relying on someone else's report (Knight, 2018). They may notice that the moment graphs appear, students stop writing.

Student work. Pull a sample of the past week's problem-set submissions and sort them by whether they show worked reasoning or only final answers.

A survey, if a wider view helps. Carefully constructed so the instructor can reflect meaningfully on their own practice and name both strengths and areas to improve.

From X to Y by Z

Walk through the evidence and let the instructor determine the goal. Then offer the structure as a scaffold for stating it clearly (Knight, 2018).

How many students show well-reasoned support for their answers? About a third. That is X. What would you want that to be?

Three quarters. That is Y. What window feels realistic? End of term. That is Z.

If they hesitate on Y, offer a range of reasonable options rather than a number. Once all three are set, reflect the whole thing back: *increase the percentage of students who show worked reasoning from 33% to 75% by the end of the term.*

Checking the goal three ways

Specific? It names an observable behavior: students demonstrating reasoning in their work.

Measurable? You can count the students showing reasoning alongside their answers.

Meaningful? Students better able to explain their reasoning perform better on assessments, and the instructor chose the target rather than receiving it.

When Someone Was Sent

The hardest version, and what to do.

Mike has taught Elementary Statistics for eighteen years and sees himself as a rigorous mathematician. His pass rates held steady until the department moved gateway courses to a co-requisite model and underprepared students entered his sections.

When his rates slipped, the chair encouraged him to meet with you, since you work on accessible design. He arrives skeptical. He believes students are simply weaker now and that making mathematics accessible is dumbing it down.

Two things are at risk. His self-concept as an instructor who teaches real mathematics, and his autonomy, because he did not choose to be here (Knowles, 1984). The chair sent him, so he may reasonably suspect an evaluation in disguise.

The moves, in order

Return his agency and establish safety. “The chair set this up, but this is not an evaluation. I am reporting nothing back, and whether we work together is your call.” Inviting him to decide whether he even stays is the best chance at keeping him open (Knight, 2007).

Acknowledge his strengths. Tell him his standards are not the problem and are in fact worth protecting. This lets him rest easy that the goal is not to water down his course.

Do not explain his data to him. Let him make sense of his own rates and connect them to topics co-requisite students typically struggle with. His read of his classroom carries more weight than any theory you brought (Knight, 2007), and it keeps the conversation on evidence rather than on him.

Offer, do not prescribe. If he is open to it, suggest a scaffolded problem designed to raise the floor rather than lower the ceiling, and say explicitly that he is free to keep, edit, or refuse it. If he continues, he names the goal. You do not assign it, because he will not pursue a goal that was prescribed (Merriam, 2001).

Rebuilding safety and self-direction

Mike begins vulnerable. You remove the threat entirely: no report to the chair, open agreement with his commitment to rigor, and his own data spoken to in his own words (Knowles, 1984). Self-direction is rebuilt by keeping every decision his. Make room for that and resist filling the space yourself (Merriam, 2001).

What success looks like

Not whether Mike redesigns his course. A quick statement of compliance may actually mean he has no intention of taking the conversation seriously. Success is his willingness to keep talking, his self-concept intact, and even a slight shift from “students are weaker” toward curiosity about how else he might engage them. If he leaves intending to try his own version of something you discussed, that is a strong win. If nothing is adopted, there is still success in the relationship changing from an audit into two colleagues investigating a shared problem. That is where the next work happens.

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PART THREE

Finding the Real Problem

Five Sources

No single one tells the truth alone.

This assessment is built for a peer-to-peer context where nobody holds authority over anyone else. Its purpose is to surface what colleagues already know and do well (Northwest Center for Public Health Practice, 2014), because an adult's prior experience is the primary resource for their learning (Knowles, 1980). Each source is designed to reveal both the assets people bring and the assumptions that may support or obstruct new learning (Mezirow, 1997).

1 • Structured interviews

One-on-one conversations with three to five instructors form the qualitative core. Tell people plainly that the purpose is not to evaluate but to understand their current methods and professional thinking. Questions are written to surface teaching philosophy, design assumptions, and current responses to both performance gaps. Ask them neutrally, without signaling which answers you are hoping for.

2 • Classroom observation

A single observation produces evidence self-reporting cannot. Record the ratio of instructor talk to student talk, the ratio of instructional to non-instructional time, the types of questions asked and whether they require recall or mathematical reasoning, and the degree to which students are active participants rather than passive recipients.

3 • Artifact review

Syllabi, Canvas pages, and sample assessments, reviewed with the instructor's permission. Look at how learning outcomes are framed, what variety of assessment types is present, whether assessments require demonstrated reasoning, and what digital tools and engagement strategies are already in use.

4 • Department-level DFW data

Individual instructor rates are usually unavailable; departmental rates are often circulated to all faculty each semester. This source anchors the current-state analysis in verifiable institutional evidence, lets performance-gap claims be confirmed independently, and helps distinguish instructional design problems from broader structural factors.

One discipline about this source. DFW data enters as a screening signal paired with completion and participation data, never as a verdict on its own. Course-level outcomes stay outside claims about any single page or assignment.

AT A GLANCE

Minimum evidence set

One conversation source, one artifact or observation, and one outcome or participation source.

Triangulation rule

No interview, survey, observation, or DFW rate stands as a verdict by itself.

Bias check

Power, nonresponse, technology confidence, and course-level versus instructor-level data.

Output

An observable performance gap stated alongside existing strengths and constraints.

Six-week rhythm

Reach, listen, observe, synthesize, then return the interpretation for correction.

The assessment should leave participants more accurately understood, not merely measured.

5 • Anonymous survey

Extends reach to instructors who will not sit for an interview or observation, and provides baseline frequency data on current practice and technical capacity across a larger group. Anonymity produces more candid answers on topics people find sensitive.

The Interview Questions

Use as written, or cut to fit your time.

Tell me about a class session that you felt went really well. What made it work?

When a student gets stuck on the same concept for the third time, what do you typically do?

What does your Canvas course currently look like from a student's perspective?

How are you currently thinking about AI in your course?

How do you think students are using AI in your course? Have any adjustments to the curriculum been made to account for it?

What would make you feel confident enough to redesign part of this course?

These reveal what an instructor values, how they understand their role, and whether AI is being treated as a design problem or a policy problem.

The Survey Questions

Anonymous, and short enough that people finish it.

In a typical week, how would you describe the primary mode of delivery in your course?

How often do you use in-class activities that require students to produce or demonstrate mathematical reasoning rather than observe it?

How many formats do you use for assessing students?

How would you describe the current state of your Canvas course from a student's perspective?

Do you currently have recorded lecture videos available for students outside of class time?

How comfortable are you with modifying the design and layout of your Canvas course?

To your knowledge, how are students currently using AI tools in your course?

How have you addressed AI use in your syllabus or assignment design?

How comfortable are you with an assessment that requires students to obtain an AI result and then expand on it to demonstrate mathematical reasoning?

Put the questions people are least comfortable answering here, not in the interview.

A Six-Week Timeline

Long enough to be real, short enough to finish.

Weeks 1–2. Outreach, scheduling, and relationship-building. Week 1 in particular includes distributing the survey and collecting departmental DFW data.

Weeks 3–4. Conducting interviews, completing observations, and collecting artifacts.

Weeks 5–6. Analysis and interview follow-up, with synthesis complete by the end of week six.

Honoring What People Bring

The process itself either respects prior experience or it does not.

Position instructors as the experts on their own practice (Knowles, 1980). Reflect interview findings back to participants to confirm you understood them correctly. Present all data as descriptive rather than evaluative. Keep artifact review and survey participation voluntary, and be fully transparent about how the data will and will not be used.

A needs assessment is not something a coach does to a professional. It is something built together (Knight, 2018).

What You Will Probably Find

The gaps are usually held in place by an interpretation, not a skill deficit.

Where the first gap is visible

Instruction organized around content delivery rather than engagement. In a class session it looks like continuous lecture punctuated by a single practice problem, leaving little time for students to wrestle with new content. It is also visible in obvious disengagement: nobody taking notes, phones out, headphones in.

Outside class it looks like plain, unstyled Canvas pages with little visible structure or visual hierarchy. Modules may follow a logical sequence that nobody can perceive. In-class materials often exist in no online form at all, which shuts out students who miss class, study independently, or need more than one way into the same content. For non-traditional students working full time, that is an immediate equity problem, not a convenience one.

This is not a platform literacy problem. Most instructors navigate Canvas comfortably. What is missing is a design for using it to engage students rather than to store material for them.

Where the second gap is visible

Less in a classroom observation, more in the details of assignment design. Many assessments rely on a numeric final answer as the primary evidence of learning. Problems built that way are easily solved by a model, which means results no longer indicate learning.

Most instructors have acknowledged AI and many have a syllabus statement prohibiting it. The deeper gap sits in the assessment design itself. Coursework rarely uses AI to help students learn – as a starting point to extend, verify, personalize, or critique. Tasks like that do not shun the tool; they require students to think with it.

The Lenses Holding the Gaps in Place

Prior experience can ground effective teaching or obstruct new learning (Merriam & Bierema, 2014).

The gaps above are rarely caused by lack of skill. They are held in place by the lens through which an instructor views their context, and that lens filters out evidence pointing toward a design they do not already use (Mezirow, 1997).

Outcomes belong to the student. A struggling or disengaged student is read as underprepared or unmotivated rather than as evidence the course is not built to meet them.

Rigor means difficulty. Redesigning for accessibility or engagement therefore reads as lowering the standard rather than meeting learners where they are.

Active learning does not transfer to mathematics. Procedural fluency is assumed to come only from repetition and independent practice.

Together these make the first gap difficult to see (Mezirow, 1997). An instructor operating inside any of them reads a lecture-heavy, minimally accessible course not as a design choice with room to improve but as an appropriately rigorous environment.

A fourth lens, about the job itself. That the work is to present mathematics clearly, and what a student does afterward is their responsibility. Under that definition, rebuilding pages, producing digital materials, restructuring the classroom, and worrying whether anyone reads announcements are simply not the job.

And one about AI. That it is an institutional problem rather than an instructor's. A syllabus prohibition gets written and the matter feels settled, while whether the assessments still function goes unexamined.

None of these are unreasonable positions, and the strengths named earlier are resources to build on rather than obstacles to remove. An instructor who has spent years learning where students get stuck, and who already knows their way around Canvas, has a strong foundation for a student-centered redesign. That experience is not absent. It has simply not yet been connected to these design questions.

Care is warranted, because instructors differ. Some hold these lenses strongly, others barely, and some already practice elements of learner-centered design without naming it that way. It is also true that many students arrive with uneven preparation, and that structural factors affect outcomes well beyond what any redesign can reach. Distinguishing what instructional design can change from what it cannot is itself part of the work.

Doing It Fairly

Two things that will distort your findings if you ignore them.

Power, even between peers

Even without supervisory authority, an assessment run by a colleague can feel threatening, because it can be read as connected to how someone is perceived in the department. The risk is sharpest for adjuncts and newly hired faculty, whose continued employment depends on contract renewal and who may reasonably worry that admitting certain practices could reflect badly on them, regardless of who is asking or why.

What the design does about it. The most sensitive questions are reserved for the anonymous survey rather than the interview. Interviews open with an explicit statement that the conversation is meant to understand current practice, not evaluate it. All findings are shared back with participants so they can correct a misreading or add context before conclusions are drawn. Interviews and observations are scheduled around instructors' existing commitments rather than added on top of them, which matters for adjuncts especially, since asking for extra time from someone stretched across multiple institutions is itself a form of pressure.

What happens without it. Sanitized answers, crafted to be socially acceptable and unrelated to anyone's actual practice.

Uneven comfort with technology

This audience spans early-career to veteran, and comfort with digital tools varies accordingly. The survey is itself a digital instrument, so relying on it as the primary source would underrepresent exactly the instructors least comfortable with technology.

What the design does about it. Interviews and observation stay the anchor sources, and neither asks any particular technical skill of the participant. The survey adds reach and anonymity, but it is not the only way in. Meaningfully engaging adults' prior experience requires more than one mode of investigation (Dillard et al., 2024).

What happens without it. The assessment misses the people whose perspective would improve the work most.

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PART FOUR

The Three Sessions

How These Are Built

Backward design for the structure, dialogue for the entry point.

Each session is planned from the instructor performance you want and the evidence that would demonstrate it, and only then from activities (Mager, 1997; Wiggins & McTighe, 2005). That is what keeps outcome, activity, and assessment pointed at the same target.

But backward design alone risks presenting as an expert-driven program, and this audience will politely ignore one. So every session opens from a problem participants have already acknowledged as their own, and the learning proceeds as dialogue among peers rather than dictation from expert to trainee (Vella, 2002).

Dialogue education alone would lack the rigor of evidence-based, outcome-driven learning. The hybrid fits an audience of experts who are not required to attend, who value their autonomy, who cannot be moved by mandates, and who engage only when they see the work solving a real problem they own.

What the research base asks of any design

Professional development that changes practice is content-focused, incorporates active learning, supports collaboration, uses models and modeling, provides coaching and expert support, offers feedback and reflection, and is sustained over time (Darling-Hammond et al., 2017). Job-embedded, coached, collaborative learning sustained over time produces change. One-time sit-and-get events rarely do (Nishimura, 2014).

Each choice here connects to that. Sessions are job-embedded and active because they work on a participant's own artifact. They are collaborative because participants pair up to redesign and give each other feedback. They are model-driven because the facilitator performs the redesign on their own materials first (Knight, 2018). And the sequence is sustained by a between-session commitment to deploy what was built, so the learning returns to be reflected on rather than ending when a session does (Darling-Hammond et al., 2017).

The shape, every time. The problem is named before any tool appears. The standard is established by experience rather than explanation. The instructor works on an artifact from their own course. The work is tested against a measure that can actually fail. Each closes with a dated commitment.

The four accessibility criteria

Referenced throughout Sessions 2 and 3. Heading hierarchy. Color contrast. Alternative text. Canvas-ready inline styling.

AT A GLANCE

Session 1

Redesign one assessment so credit depends on visible mathematical reasoning.

Session 2

Rebuild one Canvas page so students can navigate it quickly and accessibly.

Session 3

Storyboard and begin one startable resource for a commonly missed topic.

Shared design

Current course artifact, active production, peer review, reflection, and a dated next action.

Continuation

The PLC and follow-up cycle provide the sustained duration a single session cannot.

Effective professional learning is content-focused, active, collaborative, modeled, coached, reflective, and sustained.

Delivery

Every session runs face-to-face with a live remote option, scheduled inside the workday. Afterward the recording, a one-page guide, and the before-and-after examples go to a shared course site. The remote option and asynchronous materials exist for the portion of this audience the equity analysis identified, and they model the practice the sequence teaches.

Contents

SESSION ONE • 75 MINUTES

The Assignment AI Aced

Bring the assignment you have stopped trusting.

Session frame. Seventy-five minutes, delivered in person with a live remote option and materials posted afterward. Designed for six to twelve instructors working in pairs. Assumes no prior session in the sequence. A single-participant adaptation is noted at the end.

Anchoring problem

Every instructor in the room has watched a student earn full marks on homework and then fail the exam covering the same material. Most have privately concluded the homework stopped measuring anything, and have not said so out loud because the alternative is rewriting a problem set they have used for years. The cost is absorbed quietly: exam scores that surprise no one, office hours spent on material the gradebook says was mastered, and a growing suspicion that the course's early signals are worthless. The session opens on an item each participant already distrusts and stays anchored to it.

Outcome

Given one assessment item from a course they are currently teaching and a choice of two redesign moves, each instructor will redesign the item so that earning credit requires demonstrated mathematical reasoning rather than a final numeric answer, producing one revised item ready to deploy in their current or upcoming term.

Facilitation plan

0:00–0:08 Problem before tool. Each participant puts one assessment item on the table, chosen by a single instruction: bring the one you have stopped trusting. In pairs, they describe what that item is supposed to measure and what it has actually been measuring this term. No technology is introduced, and no one is asked to defend the item.

0:08–0:20 The item fails, twice. Take one of your own homework problems, paste it into a general AI model on screen, and let the room watch it earn full credit in seconds. Then apply the fix everyone has already tried: add *show all work* to the prompt and run it again. The model produces clean, plausible, fully worked steps. This second run is the segment that earns the rest of the session, because it establishes that the obvious repair is cosmetic.

0:20–0:32 Live redesign, and the model failing. Redesign the same problem on screen using the lighter move, then run the model against the redesigned version so the room watches it fail. Keep both versions visible side by side with the model's output under each. This is the before-and-after pair in its assessment form: the difference is shown rather than asserted.

0:32–0:40 Naming what blocked it. Pairs name what changed between the two versions. Take their language first, then attach the two moves to it.

Lighter move – the error hunt. The student receives an AI-generated solution containing one embedded error, locates it, explains why it is wrong, and produces correct reasoning. Credit attaches to the diagnosis, never to the final value.

AT A GLANCE

Bring

One assessment item you no longer trust.

Leave with

One revised item, tested live against a general AI model.

Evidence worth requiring

Reasoning steps, decisions, error diagnosis, interpretation, constraints, or reflection.

Watch for

Treating detection or surveillance as a substitute for valid assessment design.

Solo version

Guided redesign over screen share; the adversarial test becomes two-way.

The question is not whether AI was used. The question is whether the task still reveals the learning it claims to measure.

Heavier move – the two-stage task. The student uses AI to obtain a result, then does something with it the prompt alone cannot produce: translating an algebraic result into a graphical interpretation, testing it against a stated real constraint, or feeding it as input to a second problem.

Both are options rather than a sequence, so an instructor who wants a small change and one who wants to rebuild the item both leave with a win.

0:40–1:00 Paired redesign of their own item. Pairs rework a problem from their own course, choosing whichever move fits. Your role is coaching, not demonstration. Circulate, ask what the student would have to do that the model cannot, and stay out of the writing. Instructors who want the heavier move and stall get moved to the lighter one without ceremony.

1:00–1:10 The adversarial test. Each pair posts its redesigned item to a shared board. Every posted item is run through the AI model live, in front of the room, against two questions: would a pasted model answer still earn credit, and where exactly is the student forced to reason. An item that fails either question is coached on the spot and reposted. The test is public and the failures are ordinary, which is what keeps them survivable.

1:10–1:15 Reflection and commitment. In a shared digital reflection log, each instructor records what worked, what they will try, the specific problem and course, the date it will be assigned, and what questions remain.

How participants engage the problem through the technology

The technology is a general AI model, used adversarially rather than productively. Participants operate it themselves, against their own items, to test their own redesigns. It reveals the problem in segment 2 and certifies the fix in segment 6, and apart from the two short demonstrations in segments 2 and 3, the session never positions anyone as an audience. The shared board carries the artifacts and the timing pressure that make segment 6 public.

Session 1 is the inverse of Session 2. There the model helps build something; here it is the thing being designed against.

Instructors who arrive treating AI as a policy problem leave having used it as a design instrument, which is the belief the session exists to move.

Reflection and assessment

The adversarial test is the formative assessment, measured directly against the outcome: an item survives or it does not, judged by the same model the students will use. The closing log addresses what worked, what the participant will try, and what questions remain, and becomes the first data artifact for the monitoring cycle.

Engagement audit

Audited segment by segment. Segments 1, 4, 5, and 7 are built on the positive side of the engagement checklist (Northwest Center for Public Health Practice, 2014), since each asks participants to try activities, add information, or question a partner rather than sit. Segments 2 and 3 carry the highest disengagement risk because they are the only stretches where the facilitator holds the floor, which is why each is capped at twelve minutes and segment 2 is split across two demonstrations rather than one continuous talk. Segment 6 converts the room from observers into evaluators, the strongest engagement position the session offers.

Facilitation notes

Predicted timing failure. Segment 2 will run long. The demonstration invites war stories, and every instructor in the room has one. The stories are useful and they are not what the segment is for. Take two, name them as evidence that the problem is shared, and move.

Exam security. Instructors will ask whether pasting an unreleased item into a public model is safe. It is a fair question and deserves a direct answer rather than reassurance. The standing condition covers student names, work, and grades, none of which are involved here, but an unreleased exam item is the instructor's own risk to weigh. Offer the alternative of testing with a structurally identical stand-in.

Single-participant adaptation. Segment 5 becomes a guided redesign over screen share, and segment 6 becomes a two-way test in which the facilitator brings an item of their own and both are run.

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SESSION TWO • 75 MINUTES

Your Page, Rebuilt

Bring the page your students skip.

Session frame. Seventy-five minutes, delivered in person with a live remote option and materials posted afterward. Designed for six to twelve instructors working in pairs. Assumes no prior session in the sequence. A single-participant adaptation is noted at the end.

Anchoring problem

Every instructor in the room writes careful announcements that go unread.

The information is there, correctly stated, several paragraphs deep, and students still arrive at office hours asking what the page already told them.

Assignments come in late. An exam date gets missed by someone who was never trying to miss it. The instructor absorbs that cost quietly and usually concludes the students are not reading, which is true and is not the useful part. The session opens on the actual page each participant's students skip and stays anchored to it.

Outcome

Given one existing Canvas page from their own course, the prompt frame, and the accessibility checklist, each instructor will rebuild the page with clear, accessible structure meeting all four criteria: heading hierarchy, color contrast, alternative text, and Canvas-ready inline styling.

Facilitation plan

0:00–0:08 Problem before tool. Each participant opens one page from a course they are currently teaching, chosen by a single instruction: bring the one your students skip. In pairs, they describe what students miss on that page and what those misses have cost this term. No technology is introduced and nobody defends their page.

0:08–0:20 Establishing the standard. Using the before-and-after pair on page 23, participants cold-navigate both versions against a timer. Three specified items on the original page, then the same three on the rebuilt version. Everyone does this themselves rather than watching. The gap between the two runs is the argument. Only after the timer stops do the four criteria go on screen.

0:20–0:32 Demonstration rebuild. Describe a plain page's intended design in ordinary English, run the description through the prompt frame in a general AI model, paste the output into the Rich Content Editor, and show what survives. Then verify the result on screen against the four criteria, deliberately surfacing at least one flaw the model produced. Human verification gets demonstrated rather than asserted, and participants see the tool fail in a recoverable way before they touch it.

0:32–0:57 Guided rebuild. Participants rebuild the page they brought, entering at whichever tier fits: Canvas's built-in heading styles, a pre-built template block they paste and edit, or a design generated with the prompt frame. No tier requires

AT A GLANCE

Bring

One Canvas page students regularly miss or misread.

Leave with

One rebuilt page meeting the accessibility and navigation standard.

Quick scan

Heading sequence, descriptive links, meaningful alternative text, sufficient contrast, logical reading order, and Student View.

Use color carefully

Color may reinforce meaning, but should not be the only signal.

Watch for

The demonstration taking time away from participants rebuilding their own pages.

A page is not verified because an AI or checker says it is. Save it, view it as a student, and test the route a student must take.

writing HTML. Pairs debug together and you circulate. Your role is coaching, and the longest stretch of the session belongs to them.

0:57–1:08 Partner swap test. Partners exchange rebuilt pages and each locates three specified items on the other's page in under one minute, marking every point of hesitation. Stalls are revised immediately against the checklist. If both partners fail the same item, re-model that portion of the build for the room, since two failures on one point is a signal about the demonstration rather than about them.

1:08–1:15 Reflection and commitment. In a shared digital reflection log, each instructor records what worked, what they will try, the specific page and the date it goes live, and what questions remain.

What the Tiers Actually Produce

The same announcement, three times. Nothing added, nothing cut, only structure changed.

Below is one real Canvas announcement as an instructor typed it, then the same information with built-in heading styles and with a fuller visual hierarchy. Read the set as an argument for the tools: a student looking for the exam date should locate it faster, and that can be tested in under a minute.

BEFORE • PLAIN TEXT	TIER 1 • BUILT-IN STRUCTURE	TIER 3 • PROMPT FRAME
Welcome back! This week we're moving into Module 6. Please note that Assignment 6 is due Friday by 11:59 pm, and Quiz 6 opens Wednesday morning and closes Thursday night. Also, Exam 2 is coming up on the 24th and will cover Modules 4 through 6, so it's a good idea to start reviewing now rather than waiting until the last minute. Let me know if you have any questions!	Module 6 Overview Welcome back! This week we're moving into Module 6. Due this week • Assignment 6 – due Friday by 11:59 pm • Quiz 6 – opens Wednesday morning, closes Thursday night Coming up • Exam 2 – the 24th, covering Modules 4 through 6 • Start reviewing now rather than waiting until the last minute Questions? Let me know if you have any!	Module 6 Overview Welcome back! This week we're moving into Module 6. DUE THIS WEEK Assignment 6 Friday by 11:59 pm Quiz 6 Opens Wednesday morning, closes Thursday night COMING UP Exam 2 The 24th, covering Modules 4 through 6 Start reviewing now rather than waiting until the last minute. Questions? Let me know if you have any!

Tier 1. Four headings and two lists, all from the editor's own Paragraph menu. No markup was written and nothing was pasted.

Tier 3. The same content through the prompt frame, returned as Canvas-safe HTML. Tier 1 supplies structure; Tier 3 adds a more visible hierarchy. A Tier 2 template block lands between the two.

An instructor who stops at Tier 1 has met the Session 2 outcome in full. The tiers are entry points, not a ladder anyone is expected to climb.

Reflection and assessment

The swap test is the formative assessment, measured directly against the outcome: three items, one minute, hesitation marked.

The closing log addresses what worked, what the participant will try, and what questions remain, and becomes the first data artifact for the monitoring cycle.

Engagement audit

Audited segment by segment. Segments 1, 2, 4, and 5 sit on the positive side of the engagement checklist (Northwest Center for Public Health Practice, 2014), since each requires participants to try activities, question a partner, and add information rather than sit. Segment 3 is the highest disengagement risk because it is the only stretch where the facilitator holds the floor, which is why it is capped at twelve minutes. Segment 5 converts the room from builders into evaluators, the strongest engagement position the session offers.

Facilitation notes

Predicted timing failure, and a complication worth knowing. The demonstration will run long. It is a very easy temptation to want to give a few examples showing all the different ways a practitioner can approach the HTML, and because of this, segment 3 tends to run long. Every additional example costs rebuild time from segment 4, which is the segment that actually produces the artifact. Set a hard internal cue to stop talking and hand the floor over.

The complication: in delivery, the participant named that reiteration as something she valued rather than time wasted. So the fix is not simply talking less. It is moving the extra examples out of the demonstration and into the coaching in segment 4, where they arrive when someone is stuck and wants them, rather than in advance when nobody has asked yet.

Tier assignment is the participant's call, not yours. An instructor who chooses built-in heading styles and finishes has met the outcome exactly as fully as one who used the prompt frame. Watch for the impulse to upgrade someone mid-rebuild.

Single-participant adaptation. Runs at sixty minutes. Segment 4 becomes a guided rebuild over screen share, and segment 5 becomes a two-way swap in which the facilitator brings a page of their own and both are tested.

This session has been delivered. The notes above come from running it with a gateway mathematics colleague on July 17, 2026, not from anticipating how it might go.

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SESSION THREE • 60 MINUTES

The Class They Missed

Bring the topic you reteach most.

Session frame. Sixty minutes, delivered in person with a live remote option and materials posted afterward. Designed for six to twelve instructors working in pairs. Assumes no prior session in the sequence. A single-participant adaptation is noted at the end.

Anchoring problem

On any given day a meaningful share of a gateway mathematics class is absent, physically or otherwise. These are working adults with shifts, children, and commutes, and the absence is rarely a choice. What follows is familiar: the same lesson retaught three times in office hours, and a scores column that starts sliding for a student who missed one Tuesday and never found the way back in. The lecture existed once, in one room, at one hour. For the student who could not be there, nothing was left behind. The session opens on the specific topic each participant reteaches most often.

Outcome

Given one topic their students most often fail or miss in class, each instructor will plan and begin producing one asynchronous resource – a short worked-example video or a structured resource page – with the implementation plan completed in session and the resource published to their course within two weeks.

Facilitation plan

0:00–0:07 Problem before tool. Each participant names the one topic they have retaught most often this term, and in pairs describes what that reteaching has cost: hours, repeated explanations, the students who never came to ask. No technology in this segment.

0:07–0:18 The demonstration, and the failure first. Play the opening of a worked-example recording that starts three steps in, assuming the viewer was present on Tuesday. The room watches an absent student lose the thread inside twenty seconds. Then run the full workflow properly on the same example: choosing what to record, recording it, and publishing it into Canvas where a student would actually look. The failure comes first so the standard is felt before it is stated.

0:18–0:25 The entry point, and the guide. Pairs name what the second version did that the first did not. Take their language and attach the principle to it: the resource has to be startable by the weakest absent student, not the one who was already keeping up. The four-point guide goes on screen as the standard the work will be held to.

AT A GLANCE

Bring

The topic you reteach most often.

Leave with

A storyboarded resource, production started, and a visible publish date.

Startability test

One concept, a clear entry point, a short path, captions or equivalent access, and a location students will actually check.

Cold start

Can a colleague orient within the first sixty seconds without Tuesday's classroom context?

Watch for

Skipping the storyboard or polishing production beyond what the learning problem requires.

A resource that cannot be found or started has not solved the absence problem.

One concept only. • A short path, under fifteen minutes. • An entry point for every student. • A reasonable publication plan.

0:25–0:45 Storyboard, then produce. Pairs storyboard first: the concept, the entry point, the sequence of moves, the stopping place. Only then do they start recording. Your role is coaching. Instructors who want a structured resource page instead of a video build that instead, which satisfies the outcome equally.

0:45–0:54 Cold-start peer review. Partners exchange work and review against all four points. The core move is a cold start: the reviewing partner, who does not teach that topic the same way, watches or reads the first sixty seconds and names the exact moment they would have been lost. Anything that misses a point on the guide is revised in session.

0:54–1:00 Publication plan and dated commitment. Each instructor decides where the resource lives so students can actually find it, whether that is a page, a module placement, or an announcement link, and records in the shared reflection log what worked, what they will try, the publish date, and what questions remain. Partners exchange dates and hold each other to them.

How participants engage the problem through the technology

Two technologies, both in the participants' hands. The recording tool produces the resource and Canvas publishes it, and the publication decision is treated as part of the design rather than an afterthought, because a resource nobody can find has not solved the problem it was built for. Participants operate both directly. The only screens showing the facilitator's work are in segment 2.

Session 3 completes the arc the other two start. Session 1 redesigns what students are asked to produce, Session 2 redesigns what they arrive at, and Session 3 addresses what happens when they were not there at all. All three technologies have now been operated by the instructor rather than demonstrated at them.

Reflection and assessment

The peer review is the formative assessment, measured against the four-point guide, which restates the outcome in checkable terms. The closing log becomes a data artifact for the monitoring cycle. Because the outcome extends two weeks past the session, the publish date is the item the monitoring cycle follows.

Engagement audit

Audited segment by segment. Segments 1, 3, 4, and 5 sit on the positive side of the engagement checklist (Northwest Center for Public Health Practice, 2014), with participants trying activities, adding information, and questioning a partner. Segment 2 carries the disengagement risk as the only stretch where the facilitator holds the floor, capped at eleven minutes and broken by the deliberate failure so the room is diagnosing rather than watching. Segment 5 is the strongest engagement position in the session, since the reviewer is genuinely encountering the material cold rather than performing a review.

Facilitation notes

Predicted timing failure. Instructors will want to skip the storyboard and hit record. Recordings made without one reliably fail two points of the guide at once, running long and covering three concepts instead of one, and re-recording costs more time than storyboarding would have. Hold the first several minutes of segment 4 for planning only.

The two-week gap is where this session dies. It is the only session whose outcome is not complete when people leave the room, and good intentions do not survive a Tuesday. The partner-held publish date is the mechanism, not a nicety. Make the exchange explicit and make both dates visible in the shared log.

Watch for production perfectionism. Instructors will want to re-record for polish, and the four-point guide deliberately says nothing about production quality. A slightly rough recording that a student can start from beats a polished one that assumes they were there. Say this out loud, because otherwise the impulse costs the publish date.

Single-participant adaptation. Segment 4 becomes a guided storyboard and production over screen share. Segment 5 becomes a two-way cold start where the facilitator brings a resource on a topic outside the participant's course, and both are reviewed against the guide.

Designing for the Range

A veteran and a first-year adjunct will not take in the same explanation.

Show it more than one way

Each session presents its core practice three ways: a live demonstration on real course materials, a one-page reference guide the instructor keeps, and a before-and-after pair that shows the standard instead of describing it (Northwest Center for Public Health Practice, 2014). Nobody depends on a single format. The guide survives the session for the instructor who learns by rereading, and the posted recording serves whoever attended remotely or missed the day (Dell et al., 2015).

Give a way in at every level

These instructors are experts in mathematics and often beginners in digital design, and self-direction cannot be assumed where the skill does not exist yet (St. Clair, 2024). So every production task offers an entry at every level: Session 1's lighter and heavier moves, Session 2's three tiers running from built-in heading styles to AI-generated designs with no HTML required, and Session 3's storyboard scaffold that a first-time recorder can finish. Graduated support is what turns willingness into a finished artifact.

Make it authentic or lose them

This is a voluntary audience that cannot be compelled and that tunes out generic examples. Every session works only on something from the instructor's own current course: the assessment their own students will take, the page their own students skip, the topic their own students fail. The problem named at the opening is one the instructor already has, so relevance is recognized rather than argued for (Knowles, 1980). And choice runs through everything: which artifact to bring, which move to make, which tier to start at. For professionals whose central motivational need is autonomy, relevance and self-direction are the same design decision (CAST, 2018).

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PART FIVE

The Tools

Each technology here was selected against the same standard: it had to answer a problem instructors already face, documented in the needs assessment, rather than recommend itself through novelty (Knowles, 1980). None of the three is included for its features.

1 • Canvas Rich Content Editor, with a template library

The problem it solves

The Rich Content Editor is the page editor built into Canvas, which means every instructor already has access to it. The toolkit supplements it with a library of pre-built HTML blocks, styled inline and audited for accessibility, that paste directly into the editor and render correctly.

The problem comes straight from the assessment: course pages and announcements are well intended but dense, students skim or skip them, and essential information such as deadlines and exam dates goes unread. The cost falls back on the instructor as late submissions, missed exams, and the same question answered individually a dozen times. The barrier is not willingness. This audience is expert in mathematics and largely untrained in digital design, and no one teaching a full gateway load has the hours to learn HTML. The template library removes that requirement.

Why it fits a problem-centered orientation

The editor is never taught as a topic. Each instructor brings a page from their own course that students already skip, and the session is spent rebuilding that specific page, so the editor and the library enter only as the means of resolving the problem the instructor brought (Knowles, 1980).

What the evidence supports

The design moves the templates carry reflect the practical core of universal design for learning in online courses (CAST, 2018; Dell et al., 2015). A genuine heading hierarchy lets a screen reader navigate a page rather than deliver it as a single block. Adequate contrast stays legible on the phone screens where many working students encounter the course. A scannable structure serves the adult who opens the page late at night and needs the deadline rather than the paragraphs around it.

Access, equity, and ethics

The equity considerations favor this tool strongly. The editor is institutional and free, it runs in a browser on any machine an adjunct owns, and the tiered entry lets a veteran working with built-in styles succeed in the same session as a colleague working from the templates. The ethical safeguard operates before adoption: every block in the library is audited against the accessibility checklist before it ships, so adopting one reduces a course's accessibility exposure rather than creating it. Because the entire workflow stays inside the learning management system, no student information leaves institutional systems.

AT A GLANCE

Selection gate

Name the problem, evidence, access conditions, privacy risk, human review, and exit path before selecting the tool.

Canvas

Structures information students already need to find.

Generative AI

Drafts a design the instructor must verify and own.

Video platform

Leaves a startable resource where a one-time class meeting would otherwise disappear.

Human review loop

Define the purpose, protect sensitive information, verify the output, test it in context, and document the final decision.

A tool remains accountable only when the monitoring cycle can show whether the original problem changed.

2 • Generative AI, through a prompt frame

The problem it solves

The prompt frame is a structured, fill-in prompt an instructor copies into any current general-purpose AI model. The frame carries the technical constraints an instructor should not have to memorize: inline styles only, no document wrapper because Canvas strips one on save, headings that begin at the correct level, contrast that meets the accessibility standard (World Wide Web Consortium, 2023), and an alternative-text slot for every image. The instructor describes the intended page in plain English, and the frame constrains the output so that it survives being pasted into Canvas.

It solves the ceiling on the first technology: custom page design has historically required HTML fluency this audience does not have and should not need. It also carries the broader position that generative AI is a design pathway for instructors, not only an integrity risk to be policed.

Why it fits a problem-centered orientation

No session is about artificial intelligence. The prompt frame is the third tier of a graduated on-ramp, after built-in styles and pre-built blocks, and an instructor reaches it only when the page being rebuilt requires a design the lower tiers cannot produce.

The technology appears at the moment the problem requires it, and not before.

What the evidence supports

AI integration in adult education has to be grounded in andragogy, so that its value is conditional on serving learners' actual needs rather than on the technology's promise (Storey & Wagner, 2024). That condition is the selection standard applied throughout this part.

Four conditions on its use

Data privacy. One rule is printed on the frame itself: no student names, student work, or grades are ever entered into a public model (National Institute of Standards and Technology [NIST], 2024; Storey & Wagner, 2024).

Verification. Responsibility for the artifact stays with the instructor, because a model's statement that a page is accessible is not evidence that it is (NIST, 2024). The accessibility checklist verifies what the prompt requested, and the instructor owns what is published.

Cost. The frame is written to function on free tiers, so no instructor needs a paid account.

Inclusion. The pre-built block tier ensures that an instructor who cannot or will not use AI is never excluded from the redesign work.

3 • A licensed video platform

The problem it solves

An institutional video platform covers the full production cycle of a short instructional video: recording, automatic captioning, hosting, and embedding directly into a Canvas page. The problem is documented in the assessment. Gateway sections enroll large numbers of working adults, and on any given day some portion of them are absent, physically or in attention. The instructor reteaches the same material individually in office hours while absent students fall further behind. A five-minute worked example, produced once and embedded where students already work, addresses that absence at scale.

Why it fits a problem-centered orientation

Each instructor's video begins from a problem they already have: the single topic their students most often fail or most often miss class for. That problem selects the content of the resource, and the platform serves as the means of producing and delivering it. The tool follows the problem rather than defining it.

What the evidence supports

Posted recordings and structured asynchronous materials are among the supports the universal design literature recommends for learners who cannot attend the live event (Dell et al., 2015). The same principle governs this toolkit's own delivery, in which sessions are recorded and posted so that an instructor who missed one can still enter the sequence. The professional learning models the practice it teaches.

Access, equity, and ethics

Institutional licensing means the platform costs the instructor and the student nothing. Automatic captioning establishes an accessibility floor, with one caveat stated directly: automatic captions are a draft, and a human reviews them before publication, because mathematical vocabulary is precisely what captioning engines most often mishandle. Hosting stays inside university systems rather than on a public platform, which keeps student-facing material within the institution's privacy boundary. And a published resource persists across terms, which matters most for adjuncts, who have the least paid time to produce anything twice.

What These Tools Actually Do

An honest read, using SAMR.

SAMR's four levels group into two families (Romrell et al., 2014). Substitution and augmentation are enhancement, in which an existing task is performed somewhat better.

Modification and redefinition are transformation, in which the task itself changes. The useful question SAMR forces is whether a given use transforms the learning or reproduces an analog activity in digital form.

The template library sits at augmentation

A rebuilt page delivers the same information the plain page delivered, so the baseline move is substitution. What raises it to augmentation is a functional improvement that can be measured: a heading structure a screen reader can navigate, contrast that stays legible on a phone, and an organization a student can search rather than skim. The session's cold-navigation test – three specified items located in under one minute – exists to measure that improvement directly, so the level claim rests on an observable standard rather than an impression.

The upgrade pathway moves the page from something students read to something students act on. Embedding a short worked example with a comprehension check, or a brief self-check completed before class, redesigns the task the page performs, which constitutes modification (Romrell et al., 2014). The template library is designed so that the embed is a paste rather than a project.

The prompt frame sits at modification

Here the learner is the instructor, and the task that changes is production. Before the frame, an instructor's options were to acquire HTML fluency or go without custom design. With the frame, the task becomes describe, generate, paste, and verify. That is a significantly redesigned task, which is the definition of modification.

It is not redefinition. Students on the receiving end of an improved page still encounter exactly that: a page. This is a transformation of logistics, not of learning, and calling it more would be dishonest.

The upgrade pathway. Extending the frame beyond static structure to generate interactive page elements with built-in feedback would begin to change what the page asks a student to do rather than just how it appears. That is the direction this work points next.

Knowing Whether It Worked

Monitoring is what keeps a technology accountable to the problem it was chosen for.

After a goal is set and a practice is learned, progress gets monitored through an ongoing cycle of practice, reflection, data gathering, and adjustment (Knight, 2018). Without that discipline, tools become what technology too often becomes in institutional rollouts: equipment without a problem.

Four data sources, each with a defined purpose

The digital reflection log, completed at session close, records the instructor's commitments in their own words.

The rebuilt artifact itself, evaluated against the four accessibility criteria, establishes whether the work meets the standard.

A repeat of the cold-navigation test on the live page after it ships to students, using the same three-items-under-one-minute measure, establishes whether the improvement held outside the workshop.

Canvas page analytics, collected only if the instructor volunteers them and never read in isolation, because page views carry little meaning until they sit beside participation signals such as fewer “where do I find it” emails and on-time submission of the work the page announces.

Course-level outcome data stays outside page-level claims entirely. If DFW rates enter a later conversation, they enter as a screening signal paired with completion and participation data, never as a verdict on their own.

Review frequency

The schedule is anchored to artifacts rather than to the academic calendar, because instructors publish pages on their own timelines and a summer rebuild warrants the same attention as a fall one. The first joint review is a fifteen-minute conversation within two weeks of the rebuilt page going live. Subsequent check-ins follow the rhythm the instructor chooses. That is a deliberate design decision rather than an omission: participation is voluntary, and the goal being monitored belongs to the instructor, not to an administrative metric.

Naming the adjustments in advance

The data determines the adjustment, and the plan names the moves before they are needed. A failed re-navigation test returns the work to page structure. A rebuilt page that performs while neighboring pages do not extends the template library to those neighbors. Frustration at the AI tier lowers the entry point to the pre-built blocks.

The technologies operate under the same rule. A tool remains in the plan only as long as the data shows it serving the problem, and when monitoring indicates otherwise, the tool is replaced rather than the goal.

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PART SIX

Keeping It Going

The two problems will not close because a group of instructors met twice a semester for two years. A community creates the structure for sustained improvement, but lasting change requires individual instructors who have developed the habit of learning on their own, alongside and beyond any formal structure.

The Group

Purpose and focus

Built directly on the two performance gaps. These are not primarily content problems. They are instructional design problems, and they will not be solved by a workshop attended once.

The three sessions are where the work begins. Each ends with one artifact: an assessment item redesigned so credit requires demonstrated reasoning, a course page rebuilt for accessible structure, and an asynchronous resource storyboarded and started. Those artifacts are the learning goals the group carries forward. Each was produced once with a facilitator in the room, and the group is where the same moves get made again without one.

The focus stays practical: members examine their own course materials and assessment designs, identify specific places where student engagement breaks down or assessment integrity fails, and work together to redesign those elements. The goal is not a general conversation about good teaching but structured examination of real artifacts from real courses.

Membership

Voluntary and open to any instructor teaching a gateway mathematics course, regardless of rank. Lecturers, adjuncts, and tenure-track or tenured faculty are all eligible. Groups of five to seven work best for this kind of structured examination (Owen, 2014), and composition should reflect a range of experience. Early-career instructors bring fresh awareness of student expectations.

Veterans bring accumulated knowledge of where students consistently get stuck and which interventions have and have not worked over time.

No single expertise level should dominate. Cross-rank membership matters not as an exercise in equity for its own sake, but because these design questions look different depending on how long you have taught the course. A veteran who has run the same assessment format for eight years and an early-career instructor who arrived asking whether that format still works are both resources.

The group is not assigned. Participation is by invitation, and individuals opt in around a focus problem they recognize as real.

AT A GLANCE

Healthy meeting

A real artifact, a focused question, a structured protocol, evidence, and a next action between meetings.

Shared ownership

Rotating roles and reciprocal learning keep the facilitator from becoming the single point of failure.

What to measure

Participant learning, instructional practice, group culture, and student-facing outcomes (Blitz & Schulman, 2016).

Sustainability test

Can a new member enter, can the work continue after a facilitator leaves, and are artifacts reused rather than admired?

Warning sign

Conversation stays pleasant but no artifact, evidence, or practice changes between meetings.

The group is successful when it strengthens individual habits of inquiry rather than creating dependence on the group.

Meeting structure

Twice per semester, roughly ninety minutes. The frequency is intentionally modest because the real work happens between sessions, not during them. Each meeting opens with a brief review of what members tried since the last one, moves into the main protocol, and closes with explicit commitments about what each person will do before the next.

Facilitation rotates. No permanent facilitator owns the group. That is a structural choice rather than a logistical convenience: when facilitation rotates, direction is distributed rather than resting with a single person whose departure would destabilize things (Owen, 2014).

Norms are established by members in the first session rather than handed down. They typically include: bring an artifact; assume good intent when a colleague's work is examined; separate the design from the designer when offering feedback; do not attend if you are not prepared to participate.

Two Protocols

Structure is what keeps it from becoming a pleasant conversation.

A tuning protocol, for artifact review

One member brings a piece of course design – an assessment question, a Canvas module, a problem set – and poses a focusing question to the group, such as *does this question require reasoning that AI cannot replicate?* The group reviews the artifact silently, offers warm feedback on what is working and cool feedback on what is not, and then the presenter reflects. The presenter listens during feedback without defending.

This creates the productive friction that separates genuine professional learning from pleasant agreement (Owen, 2014). The friction is the point.

A structured evidence review

Members bring data from their own courses – exam item analysis, submission patterns, DFW rates on specific topics – and the group works together to ask what the evidence suggests about the instructional design behind it. This connects directly back to the needs assessment: the same diagnostic lens used to identify performance gaps is now applied to examining whether redesign is closing them.

How the Structure Builds Internal Motivation

Each choice is doing something.

Voluntary participation means nobody attends because someone recorded that they did. Self-chosen focus problems mean the group works on something its members actually care about. Rotating facilitation distributes ownership rather than concentrating it. The absence of a compliance report means the work is assessed by whether anything changes in members' courses, not by whether a form was submitted on time.

Professional learning communities significantly mediate the relationship between professional development and actual instructional integration – a finding drawn from more than sixteen thousand STEM teachers (Liu et al., 2024). The mechanism is relevant here: these communities provide a less hierarchical environment where members treat each other as like-minded colleagues, grounded in shared purpose and mutual respect. That is precisely what this design is trying to build.

Why It Works, and What Would Break It

Each strength has a matching failure mode. Naming them is not pessimism.

Autonomy

What supports it. Voluntary membership, self-chosen focus problems, rotating facilitation, and member-written norms. Nobody is assigned a group, an agenda, or a question.

The structural risk is institutional rather than internal. If leadership decides the group is valuable enough to formalize – requiring documentation, tying participation to annual evaluations, or assigning faculty to groups – the voluntary structure collapses. Externally regulated behavior is poorly maintained once the controlling contingency is removed (Niemiec & Ryan, 2009), and the inverse holds too: once a contingency is added to a voluntary structure, autonomous motivation tends to be replaced by external regulation.

The design logic needs to be shared with leadership before that pressure arrives, not in response to it.

Competence

What supports it. Work on real artifacts, visible progress, and shared expertise rather than abstract discussion. Members leave with something rather than with notes.

The structural risk is a group where one voice dominates, which turns everyone else into spectators and quietly removes the competence support the design depends on. Rotating facilitation helps. Attention to airtime helps more.

Relatedness

What supports it. Cross-rank membership, shared purpose, and a protocol that separates the design from the designer.

The structural risk is the one that cannot be designed away. An adjunct may not feel free to disagree with a tenured colleague regardless of what the norms say. Contract renewal is real and no stated norm removes it. The honest response is not a written rule but facilitation that is actively attentive to who is speaking and who is not, and a willingness to notice when the pattern is about rank rather than about ideas.

Networks Already Out There

You do not have to build these. They exist, they are open, and they are specific to this work.

AMATYC

The American Mathematical Association of Two-Year Colleges maintains a professional network and an annual conference focused on the first two years of college mathematics, where gateway courses live. Its members work at two-year colleges, but their daily work matches this audience's: people teaching College Algebra and Statistics to underprepared adults, comparing notes on what worked.

MAA

The Mathematical Association of America publishes practitioner-oriented resources on undergraduate mathematics pedagogy, including work on gateway course redesign, placement, and co-requisite models.

MAA OPEN Math

Online Professional Enhancement Network. NSF-funded workshops on active learning built specifically for higher education mathematics faculty, run online across a semester rather than as a single day. Recent offerings have covered team-based inquiry learning and alternative grading, both directly relevant to the two gaps this toolkit addresses.

MAA Project NExT

New Experiences in Teaching. A structured program for new mathematics PhDs entering their first faculty positions, pairing fellows with mentors and a cohort of peers. If you are early career, this is the highest-value single thing on this page.

Project EMBER

Eliminating Mathematics Barriers through Evidence-based Reforms. A network built expressly around teaching-focused faculty in gateway mathematics courses, organized so that institutional teams coalesce into affinity groups around the practices most relevant to them. Most members teach at research universities, but its online community is open to teaching-focused faculty elsewhere.

SIGMAAs

MAA's special interest groups, including ones on mathematics education research and on teaching advanced and introductory courses. Narrower than the parent organization and easier to find your specific question inside.

Online practitioner communities

Lower-barrier and ongoing. Forums, mailing lists, and social communities where instructors post what they tried last week and what happened. Less authoritative than a journal and far more current, which for design questions is often the more useful trade.

These matter most for adjuncts and lecturers who may not feel fully embedded in a single department's intellectual culture, and who benefit most from a professional home that does not depend on their contract being renewed.

Building the Habit Yourself

Self-directed does not mean solitary. It means the questions are yours.

Self-directed learning is a process in which learners take the primary initiative, with or without help, for planning, conducting, and evaluating their own learning (Knowles, 1980). Self-directed professionals actively seek out colleagues, networks, and resources. Self-direction is an orientation toward one's own learning, not withdrawal from other people (Abdullah et al., 2008).

Joining a network gives you somewhere to take a question. The four practices below are what generate the questions in the first place, and together they build a small community of practice whether or not a formal one exists where you work.

An instructional experimentation log

Keep a running record – a shared document, a page in a notes app, a paper journal – of instructional experiments tried between meetings. The entry does not need to be long. What you changed, why, what you noticed, and whether you would try it again.

The purpose is to build the habit of treating your own course as a source of evidence rather than a fixed routine. For instructors who have taught the same course with the same materials for years, that critical examination rarely happens without a structure that makes it visible.

A peer observation cycle

Once per semester, pairs observe one another's classes and debrief using a brief structured reflection. The observation is not evaluative. No form is submitted to a department chair. The observer watches for one specific thing the observee named in advance as something they are working on.

This extends professional learning into the actual site where the performance gap lives. It also strengthens the relational foundation that makes the tuning protocol more productive, because instructors who have seen one another teach give feedback that is more specific and better grounded.

Letting the question drive the reading

When you bring a resource to a colleague, the goal is not to assign reading. It is to introduce them to a community and let their own professional questions decide what they do with it.

A veteran who discovers that another institution solved exactly the assessment problem they have been struggling with is far more likely to act on that than on anything assigned in a workshop. The same resource, handed over two different ways, produces engagement or compliance depending entirely on who chose it.

Digital tools between meetings

Collaborative annotation lets one member's reading become a shared conversation without requiring a synchronous meeting.

Short screen recordings let instructors share walkthroughs of their Canvas pages or assessment designs and get feedback without waiting for the next session.

Measuring the Right Thing

Skill is whether you can build the page. Orientation is whether you are still asking a year later.

Measuring a learning orientation is different from measuring skill acquisition. Skill acquisition can be assessed by whether an instructor can now build a Canvas module with proper visual hierarchy. Learning orientation is measured by whether that instructor is still asking questions and trying new things twelve months on. Three measures are practical.

An annual self-assessment. At the start and end of each academic year, members complete a brief self-assessment of readiness for self-directed learning (Guglielmino, as cited in Abdullah et al., 2008), asking about openness to learning opportunities, initiative in seeking new approaches, and comfort with uncertainty in practice. Year-over-year responses show whether the orientation is developing.

Experimentation logs reviewed collectively each spring, not for the quality of the experiments but for whether instructors are continuing to run them. Someone who tried two new things in fall and four in spring is developing a learning orientation regardless of whether any individual experiment succeeded.

Conversation patterns, tracked informally by the rotating facilitator. Instructors developing a learning orientation tend to bring more questions than conclusions to meetings. A shift in that ratio is a reliable signal that something is changing in how members relate to their own practice.

Why this connects to internal motivation

Adults are most powerfully motivated by internal factors (Knowles, 1980). None of the strategies above produce compliance by requiring participation. All of them create conditions where an instructor who is genuinely curious about their students' performance has somewhere productive to take that curiosity.

There is one thing the self-directed learning literature alone does not supply: developing a self-directed orientation sometimes requires examining and revising assumptions that have organized a professional's practice for years (Mezirow, 1997). For a veteran who has used the same course design since 2010, self-direction is not primarily a skill deficit. It is a question of whether existing assumptions about what good teaching looks like can be examined from the outside. The experimentation log and the peer observation cycle are both designed to create that outside view.

What Threatens This

Three honest threats, each with a specific answer.

1 • Institutional pressure to scale and mandate

A thriving voluntary group will eventually attract attention from leadership who want to extend it to everyone. The proposal will arrive framed as a compliment. From there the path is predictable: voluntary participation becomes expected, then required, documentation gets added, and the structure that made it work is dismantled in the name of spreading it.

Sustainability is defined by a principle so embedded in culture that it survives changes in key personnel, not by universal participation (Owen, 2014).

What. In the first semester, write a brief design rationale document, no more than two pages, that names voluntariness and self-chosen focus problems as the mechanism of the design rather than optional features. Explain in plain terms why an assigned group with a predetermined agenda produces compliance rather than learning.

When. Before the end of the first year, while the group is small enough that its logic is easy to explain and before any scaling conversation begins.

Who. The facilitator and the department chair together. The chair needs to understand and be able to articulate the logic independently.

Evidence it worked. When administration proposes extending the group to all faculty, the chair can articulate why mandatory participation would undermine it. Any expansion preserves the opt-in structure. No group is assigned membership.

2 • Facilitator departure or burnout

This design came from one person, and the group initially depends on that person's understanding of why it works. A system that cannot outlast its designer has not reached sustainability regardless of how well it currently functions (Owen, 2014).

What. Rotate facilitation fully into the group by the end of the first year, so that at least two other members could facilitate a session and describe the design logic without the originator. Maintain the rationale document as a standing resource. Identify one member who can serve as primary contact for new participants.

When. Facilitation rotation begins in the first session. A second capable facilitator is identified by the midpoint of year two.

Who. All members participate in the rotation. The secondary facilitator is identified collaboratively, not appointed.

Evidence it worked. At least one session per year is facilitated by someone other than the originator. When asked to describe the group to a new colleague, at least two members can do so accurately, including the rationale for voluntary membership and self-chosen focus problems.

3 • Disengagement after the novelty period

New structures attract curious participants, and in the first year that curiosity carries momentum. By year two or three, participants who joined out of interest rather than around a genuine professional problem may disengage, and sessions can begin to feel repetitive. This is drift toward maintenance without growth rather than outright dissolution (Mulford, as cited in Owen, 2014).

What. Hold an annual spring share-out in which each group briefly presents what it worked on that year and what, if anything, changed in members' courses as a result. This is not a compliance report. It is an occasion for the group to make its work visible to the wider department, which gives the group a reason to produce something coherent across the year and lets colleagues who were not involved see what the work looked like.

When. End of each academic year, using existing department meeting infrastructure. No additional meeting time needed.

Who. All members present. The department chair or equivalent attends as audience, not evaluator.

Evidence it worked. Groups continue meeting into years two and three without external requirement. New instructors ask to join after the share-out. At least one group can point to a specific change in course materials or assessment design that resulted directly from the group's work.

Contents

PART SEVEN

What This Is Built On

The Stance

Adults keep ownership of the problem, or they do not learn.

Professional learning should not position educators as passive recipients of expertise. It should help them examine their practice, interpret evidence, name meaningful problems, and decide what to do next.

Adult professionals should participate meaningfully in decisions about their own learning (Knowles, 1980). Their prior experiences are not background information; they shape how new ideas get interpreted, how problems are understood, and whether a proposed change reads as relevant. Professional learning should therefore begin with authentic problems of practice and invite people to connect new learning to what they already know.

Coaching as a relationship between equals

Coaching should be a relationship between equals rather than a process through which an authority transfers knowledge to a subordinate (Knight, 2007). The coach brings expertise, and the educator brings expertise grounded in experience, context, and knowledge of their own students. The responsibility is to ask, listen, share what you know, and support reflection without taking over.

Equality does not require pretending both people know the same things. It requires recognizing that both hold knowledge that matters, and that decisions should emerge through dialogue rather than be imposed.

That stance carries a hard consequence: the educator has to be allowed to own the problem. Nobody should define a colleague’s problem before that person has had the opportunity to examine their own reality. A coach may contribute questions, observations, and evidence, but the educator retains ownership over how that evidence is read, what problem is named, and which goal is chosen. When a coach diagnoses prematurely, the process undermines self-concept and reduces professional learning to compliance (Knowles, 1984).

What this means for equity

Adults do not enter professional learning on equal footing. Rank and contract status shape who can afford to admit a problem in front of colleagues, and scheduling shapes who can attend at all. Treating everyone identically is therefore insufficient.

Universal design offers the more useful standard, since multiple means of engagement, representation, and expression are built in from the start rather than added later as accommodation (CAST, 2024; Dell et al., 2015).

It also requires asking whose experience is treated as authoritative, and avoiding the confusion of confidence with expertise or silence with disengagement. Autonomy is only meaningful when participants have genuine access to the learning process.

AT A GLANCE

Stance

Adults keep ownership of the problem.

Coherence line

Audience → problem → design
→ technology → evidence → continuation.

Backward check

Every tool should point back to a documented problem; every problem should point forward to an observable change.

Equity check

Whose constraints, access, risk, and professional authority are built into the design?

Final question

Would a colleague know what to do next without having taken the course that produced this toolkit?

A coherent toolkit can be traced in both directions without inventing a missing connection.

What Changed Along the Way

Some of this confirmed what I already believed. Some of it did not.

Andragogy confirmed my starting position on ownership, but working through all six principles showed me that a single principle can cut two ways (Holton et al., 2001; Knowles, 1980). Prior experience is an asset where it applies and an obstacle where expertise in one domain conceals inexperience in another.

Experiential learning confirmed a preference for cycles of action, reflection, conceptualization, and renewed experimentation (Dillard et al., 2024).

Transformative learning challenged me to move beyond asking whether a strategy worked and examine what assumptions shaped the educator's interpretation of the situation (Mezirow, 1997). The coaching cycle challenged me differently, by requiring restraint at the exact moment I most want to be useful (Knight, 2018). And self-determination theory clarified that autonomy alone is insufficient, since sustained motivation also depends on competence and relatedness (Ryan & Deci, 2000).

Two things I now do differently

I stopped trusting my own account of what a technology does. SAMR gave me a test to run on my own design, and it returned an uncomfortable answer: the tool I was most pleased with sits at augmentation, not where I would have placed it (Romrell et al., 2014). I now classify before I claim.

I stopped explaining what good design looks like. Dialogue education convinced me that adults learn from doing rather than from being told (Vella, 2002), so a session now puts two versions of a real artifact side by side and lets the difference carry the argument.

How the Parts Fit Together

One thread runs through all seven.

The instructor supplies the problem, or nothing happens.

That is why Part 1 starts with who these people are rather than what they should know. Why Part 2 is a coaching stance instead of a delivery method. Why Part 3 finds the problem before Part 4 designs anything.

Parts 3, 4, and 5 are a single line. The assessment found dense pages students skim past, so the tools include a template library and a prompt frame. It found adjuncts spread across campuses and instructors with no shared free hour, so every session runs face-to-face with a live remote option and posts its materials afterward. That is not only an equity move; it demonstrates the same accessibility principle the sessions teach.

Parts 2 and 6 are what make the rest survive. The partnership in Part 2 is what brings an instructor back, because it is coaching rather than evaluation, and it is the precondition for any of the Part 5 tooling to matter. Without that relationship, the tools are resources nobody opens. The voluntary membership, modest meeting frequency, and absence of hierarchy in Part 6 keep the learning internally motivated, which is what keeps people learning after the last session ends.

The three sessions and the group are both downstream of the same two performance gaps, and for a while they sat adjacent to one another without connecting. The group's purpose is now tied directly to the artifacts the sessions produce, which is what turns a set of good parts into a continuous line of practice.

Contents

WORKING MATERIALS

Worksheets and Reusable Tools

Everything the sessions reference, plus additional planning and evidence tools for coaching, needs assessment, redesign, and continuation.

Digital materials remain available in the repository. Open the complete Session 2 materials and HTML template library.

Use it when	Working page
You need a Canvas-safe first draft	1 · AI Prompt Frame
You are verifying a course page before publication	2 · Accessibility and Navigation Checklist
You are planning a short asynchronous resource	3 · Four-Point Resource Guide and 4 · Storyboard Scaffold
You are closing a session and setting follow-up	5 · Session Reflection
You are choosing a level of support	6 · Template Library Selection Guide
You are planning, coaching, redesigning, or sustaining the work	7–12 · Additional planning and evidence tools

WORKSHEET 1

AI Prompt Frame

Use during Session 2 when the page you need is not already available in the template library.

COURSE	PAGE	DATE
--------	------	------

You are rebuilding one Canvas page for a gateway mathematics course.

Hard requirements:

- Use a clear heading hierarchy.
- Use descriptive link text.
- Include meaningful alternative text placeholders for every image.
- Do not rely on color alone to communicate meaning.
- Use Canvas-ready inline styling only. Do not include <style>, <script>, or a full document wrapper.
- Preserve every fact, date, requirement, and link supplied below. Do not invent content.
- Return the rebuilt page first, followed by a short verification checklist.

Audience and purpose:

[Describe the students, what this page must help them do, and what they commonly miss.]

Content to rebuild:

[Paste the original page text here.]

Optional design direction:

[Name the structure or emphasis you want. Leave blank when unsure.]

Human verification notes

Record anything the model omitted, invented, misordered, or made inaccessible.

WORKSHEET 2

Accessibility and Navigation Checklist

Verify the saved page in Student View before it is published.

COURSE	PAGE	REVIEWER
--------	------	----------

- ☐ **Heading structure.** The title is followed by headings in a meaningful order; bold text is not substituting for headings.
- ☐ **Navigation.** A student can locate the three most important items quickly without reading every sentence.
- ☐ **Links.** Link text identifies the destination or action without relying on “click here.”
- ☐ **Images.** Meaningful images have concise alternative text; decorative images are marked appropriately.
- ☐ **Color and contrast.** Text remains readable and color is not the only way meaning is communicated.
- ☐ **Reading order.** Information remains meaningful when read from top to bottom and at a narrow width.
- ☐ **Keyboard and focus.** Links and controls can be reached and their focus is visible.
- ☐ **Canvas-safe construction.** The page saves, renders, and survives Student View without stripped styling.
- ☐ **Human check.** Dates, mathematics, requirements, links, and AI-generated descriptions have been verified by a person.

Three-item navigation test

ITEMS THE REVIEWER MUST FIND	WHERE HESITATION OCCURRED
------------------------------	---------------------------

Revision before publication

WORKSHEET 3

Four-Point Resource Guide

Use for any asynchronous video or structured resource page.

COURSE	TOPIC	FORMAT
---------------	--------------	---------------

Standard	Pass test	Notes
One concept only	The resource has one stated learning purpose and a deliberate stopping point.	
A short path	The resource is brief enough, under fifteen minutes, that the student who most needs it is likely to begin and finish it.	
An entry point	An absent student can orient without having been present for the class that produced the need.	
A publication plan	The resource has a named location, visible link, and publish date.	

Access check

- ☐ Captions, transcript, or equivalent text access are available.
- ☐ Visuals are described or explained in the narration.
- ☐ The resource can be reached from the course location where students will look.

WORKSHEET 4

Storyboard Scaffold

Plan the entry point, sequence, and stopping place before recording or building.

The one concept

The entry point

What does a student who missed class need before this makes sense?

Move 1

Move 2

Move 3

Where I stop – and what I deliberately do not cover

WHERE IT LIVES IN THE COURSE

PUBLISH DATE

WORKSHEET 5

Session Reflection

Complete during the final minutes of a session; the record belongs to the participant.

SESSION	ROLE	ARTIFACT WORKED ON
---------	------	--------------------

1 • What worked?

What in the session actually helped – a model, tier, prompt, peer test, or conversation?

2 • What will you try?

Name the exact page, problem, or resource and the date it will be used.

COURSE LOCATION	GOES LIVE / IS ASSIGNED ON
-----------------	----------------------------

3 • What questions remain?

Follow-up date

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WORKSHEET 6

Template Library Selection Guide

Choose the lowest level of support that solves the page problem. No tier requires writing HTML.

Open the complete Session 2 repository and template library.

Tier	Use when	What you do	Verification
1 • Canvas headings	The content is sound and primarily needs structure.	Apply Canvas heading styles and lists using the editor.	Run the three-item navigation test.
2 • Template block	A tested page pattern already matches the need.	Paste the block, replace the words, and keep the markup intact.	Run the full accessibility checklist.
3 • Prompt frame	The needed design is not represented in the library.	Describe the design in ordinary language, generate a draft, and revise it.	Verify every fact, link, style, and accessibility requirement by hand.

The problem this page must solve

TIER SELECTED

WHY THIS IS ENOUGH

WORKSHEET 7

Needs Assessment Evidence Planner

Plan a small, ethical evidence set that can reveal both strengths and performance gaps.

Performance gap under investigation

State what is not happening consistently or effectively. Avoid naming a content topic as the gap.

Source	What it can show	What it cannot show alone	Safeguard
Interview / conversation			
Observation / video			
Artifact / course page / assessment			
Outcome / participation data			
Anonymous survey			

Existing strengths to preserve

Power, access, or nonresponse concerns

How the interpretation will be returned to participants for correction

WORKSHEET 8

Instructor-Owned Goal Builder

Turn jointly interpreted evidence into a specific goal the instructor chooses.

X • CURRENT STATE	Y • DESIRED STATE	Z • TIME WINDOW
-------------------	-------------------	-----------------

Complete goal statement

Increase / decrease / improve _____ from X to Y by Z.

Evidence source

Why this matters to the instructor

- ☐ The goal names an observable behavior or outcome.
- ☐ The measure is available without creating an unreasonable burden.
- ☐ The target was selected by the instructor, not assigned by the coach.
- ☐ The goal fits what the instructor can actually change inside the course context.

FIRST ACTION	REVIEW DATE
--------------	-------------

WORKSHEET 9

Assignment Redesign Planner

Redesign one item so earning credit requires visible mathematical reasoning.

Current assessment item

What it is intended to measure

What a model can currently produce

Reasoning that must become visible

Redesign moves under consideration

- ☐ Error diagnosis and correction
- ☐ Interpretation across representations
- ☐ Application to a local or stated constraint
- ☐ Comparison of two methods or solutions
- ☐ Process annotation or decision explanation
- ☐ Staged checkpoint or brief oral follow-up
- ☐ Reflection on AI output, limits, or verification

Revised item

ADVERSARIAL TEST RESULT

WHERE THE STUDENT MUST REASON

WORKSHEET 10

Course Page Navigation Test

Test whether a student can find what matters before publishing the page.

PAGE	REVIEWER	DEVICE / VIEW
------	----------	---------------

Item to find	Time	Hesitation or wrong turn	Revision
1.			
2.			
3.			

What the reviewer expected first

What the hierarchy actually emphasized

- ☐ The page title names the purpose.
- ☐ The first screen contains an orientation or immediate next action.
- ☐ Dates and required actions are visually distinct.
- ☐ Link text makes sense outside its surrounding sentence.
- ☐ The page remains usable on a narrow screen.

Final publication decision

WORKSHEET 11

PLC Evidence Review

Keep collaborative learning centered on an artifact, evidence, and an actionable next step.

MEETING DATE	PRESENTER	ARTIFACT
--------------	-----------	----------

Focused question from the presenter

Evidence noticed

Questions before suggestions

What appears strong or promising

What may need revision or testing

NEXT ACTION	OWNER	REVIEW DATE
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WORKSHEET 12

Implementation and Improvement Log

Track whether a design change was used, what evidence appeared, and what should change next.

Date	Change tried	Why	Evidence collected	What happened	Next adjustment

Pattern emerging across entries

QUESTION TO CARRY INTO THE PLC	EVIDENCE NEEDED NEXT
--------------------------------	----------------------

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